

# Taranga CTF 2026 – Penetration Testing Style Report

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**Event:** Taranga Winter CTF 2082

**Result:** 3rd Place

**Category:** Linux Privilege Escalation

## Executive Summary

This report documents the exploitation workflow used during Taranga CTF 2026, demonstrating structured penetration testing methodology including enumeration, vulnerability identification, exploitation, and privilege escalation.

## Initial Access & Enumeration

Access was obtained to a remote Linux host via SSH running on a non-standard port. Post-authentication enumeration was conducted to identify local users, permissions, and potential escalation vectors.

## Vulnerability Identification

A custom SUID binary was discovered that executed system commands without using absolute paths. This introduced a PATH hijacking vulnerability, allowing attacker-controlled binaries to be executed.

## Privilege Escalation – PATH Hijacking

By manipulating the PATH environment variable and placing a malicious executable in a writable directory, the vulnerable SUID binary executed attacker-controlled code, resulting in escalation to an intermediate user.

## Root Escalation via Sudo Environment Abuse

Further enumeration revealed a sudo rule permitting execution of a backup script with preserved environment variables. This misconfiguration allowed execution of arbitrary commands as root.

## Impact

This vulnerability chain allowed full system compromise from a low-privileged account, demonstrating the risk of improper SUID handling and unsafe sudo configurations.

## Defensive Recommendations

Use absolute paths in privileged binaries, sanitize environment variables, restrict sudo permissions, and audit custom administrative scripts regularly.

## Conclusion

The challenge reinforced real-world Linux privilege escalation techniques and aligns closely with professional penetration testing workflows.